

Global HIV Trends & Treatment Effectiveness Dashboard

Olufunmilola Lanade | Healthcare Data Analyst | Power BI Analytics

Project Overview

This portfolio report presents a concise, public-facing version of the Power BI capstone project analysing global HIV burden, treatment access, mortality trends and public health intelligence using WHO data from 2000-2024. It is designed to showcase the dashboard output and insight quality without exposing the underlying PBIX file, full DAX, raw dataset, or Power Query steps.

Dashboard Overview

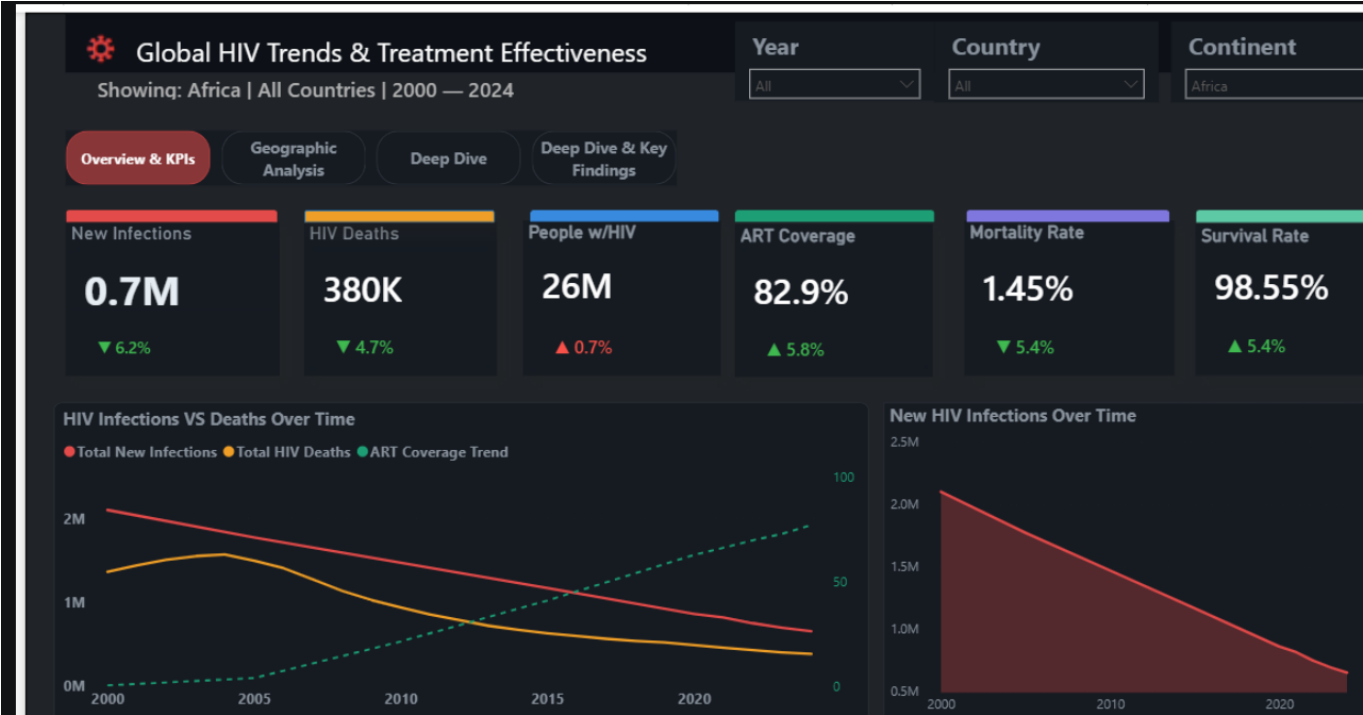


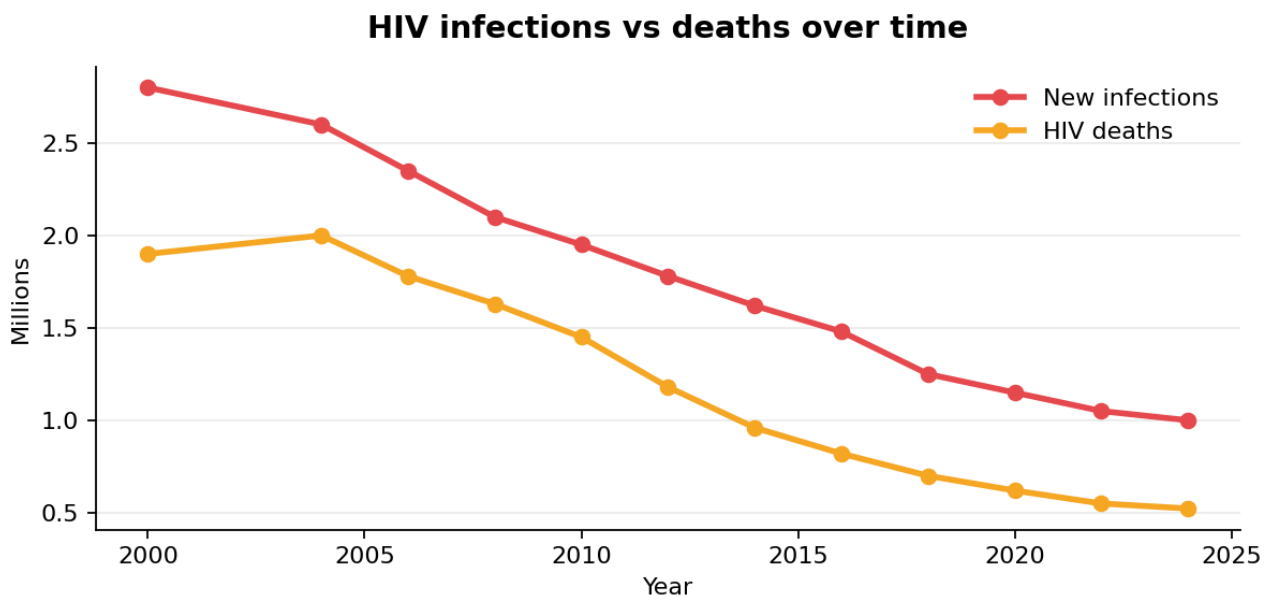
Figure 1 - Global HIV Trends & Treatment Effectiveness dashboard overview. The report focuses on KPIs, trend analysis, geographic burden and treatment effectiveness.

Key KPI Snapshot

New infections	1.0M
HIV deaths	523K
People living with HIV	35M
ART coverage	77.2%
Mortality rate	1.49%
Survival rate	98.51%

Key Insights with Visual Evidence

Infections and deaths are declining, but the epidemic is not over



The 25-year trend shows a substantial decline in both new infections and HIV-related deaths, reflecting the impact of prevention programmes and expanded access to antiretroviral therapy. However, one million new infections in 2024 still indicates a continuing prevention challenge.

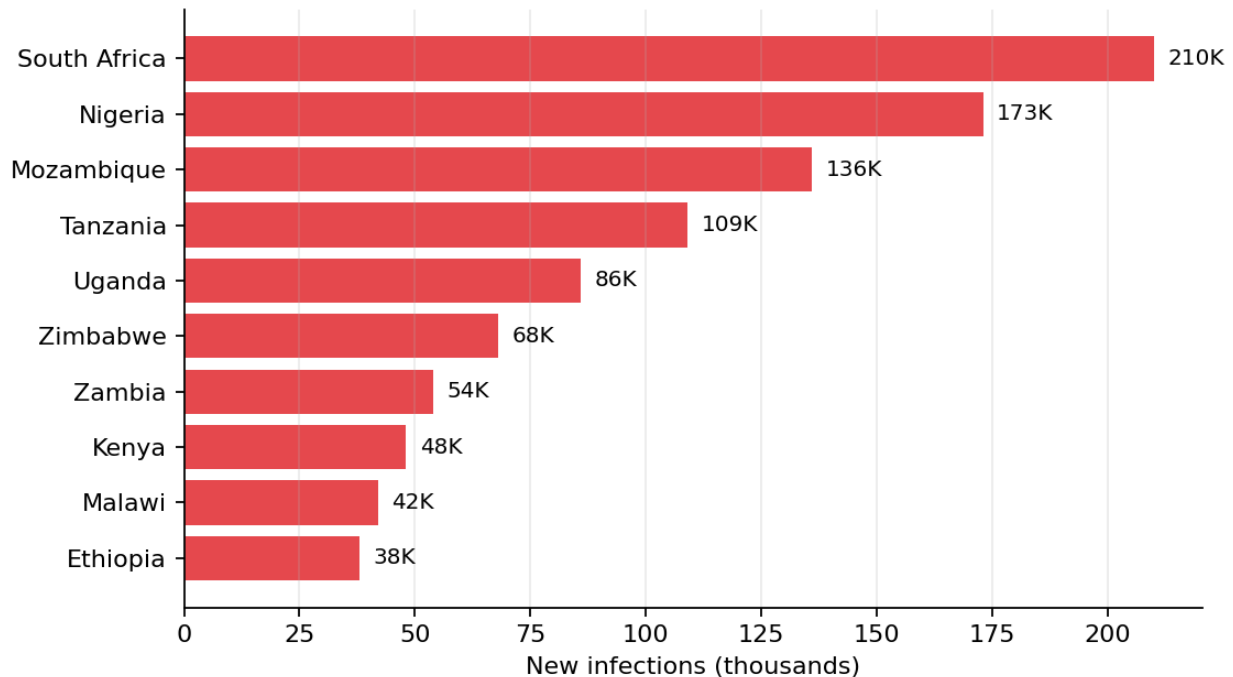
Insight caption

Deaths fell faster than infections after ART scale-up, suggesting that treatment expansion is improving survival. Continued prevention investment is still required because new infections remain high enough to sustain long-term health system burden.

Geographic Burden and Treatment Access

Sub-Saharan Africa remains the centre of the burden

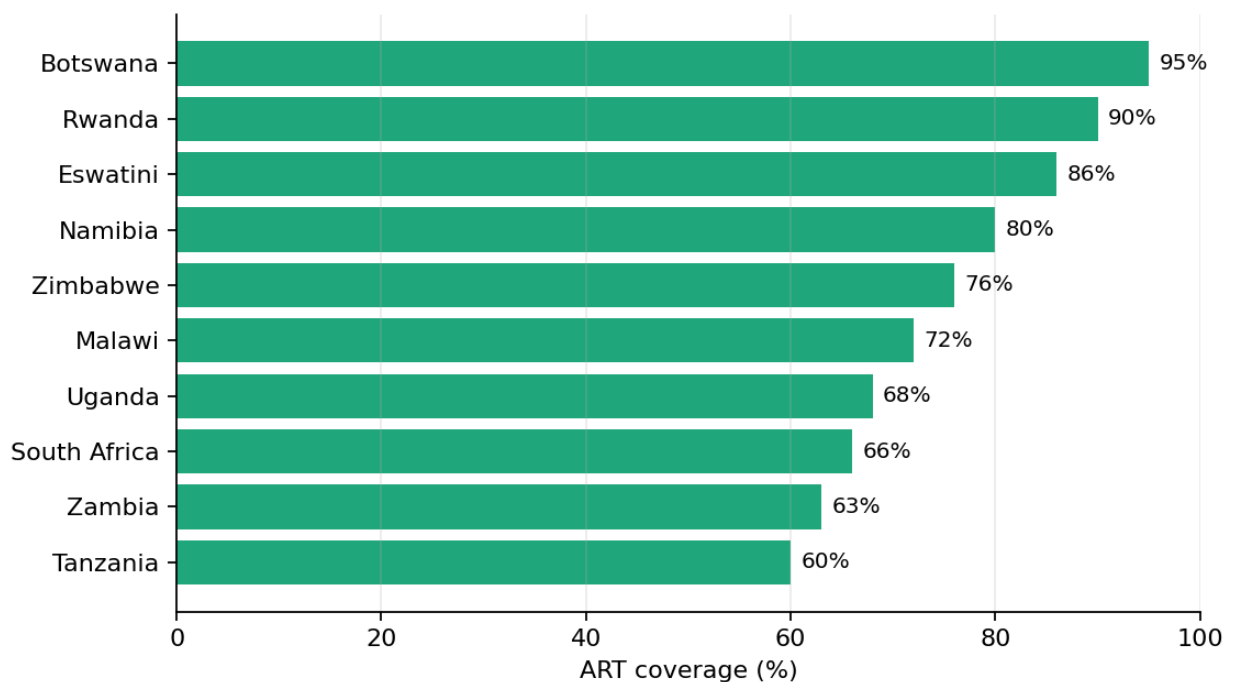
Top 10 countries by new HIV infections (2024)



The highest new infection volumes are concentrated in African countries. South Africa and Nigeria account for the largest absolute burdens, making them priority countries for prevention, testing and treatment scale-up.

Treatment access is a measurable success factor

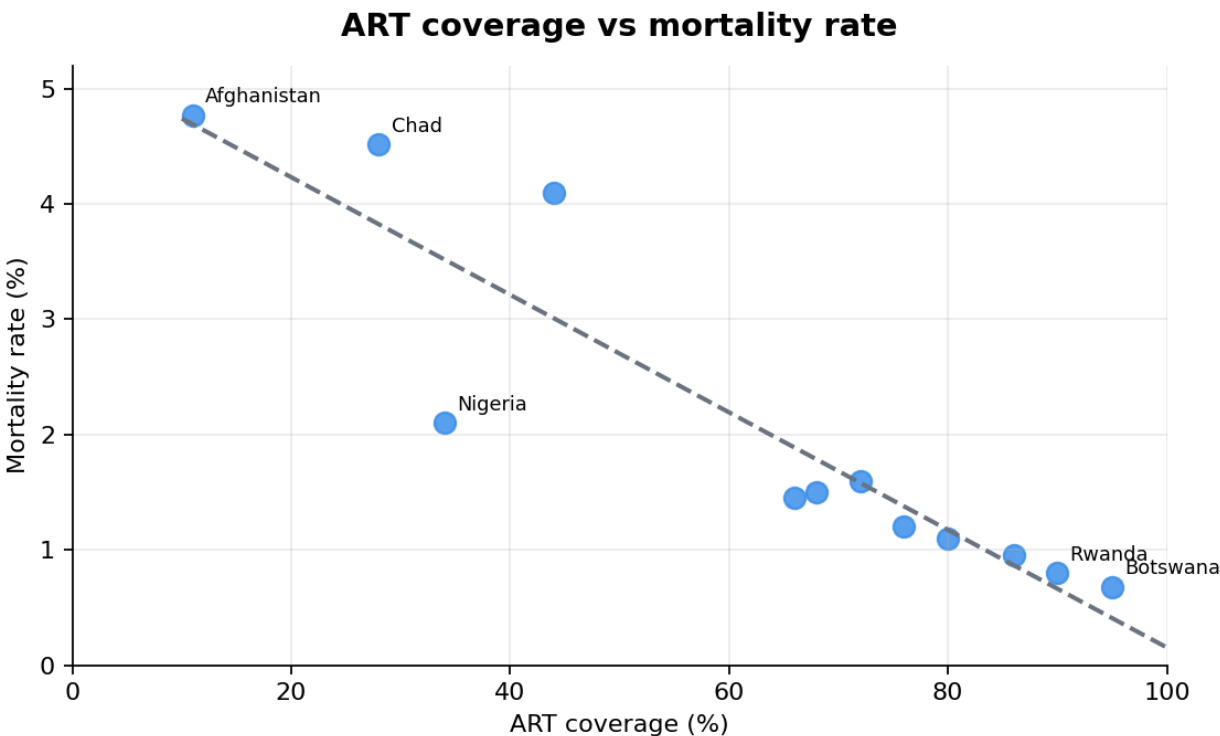
Top countries by ART coverage (2024)



High ART-performing countries such as Botswana and Rwanda demonstrate that strong treatment access is achievable in high-burden settings when systems, funding and community delivery models are aligned.

Treatment Effectiveness and Risk Segmentation

Higher ART coverage is associated with lower mortality



The downward relationship between ART coverage and mortality supports the core public health message: treatment access is not only a coverage metric but an outcome driver. Countries with low ART coverage sit in the high-mortality risk zone.

High-Risk Country Monitoring View

Country	New infections	HIV deaths	ART coverage	Risk note
Nigeria	173K	52K	34%	High burden + low treatment
Chad	16K	11K	28%	Very high mortality risk
Afghanistan	Data-limited	High risk	11%	Conflict/health access challenge
Cameroon	38K	21K	44%	Treatment gap requires action

Portfolio protection note: this summary includes outcome visuals and business insights only. It does not include the editable PBIX model, DAX source library, Power Query transformation logic or raw dataset.

Business Impact and Recommendations

Business Impact

This dashboard supports public health stakeholders by translating multi-country HIV data into a clear intelligence product. It helps identify where prevention investment should be prioritised, where treatment access is underperforming, and which countries require urgent monitoring because of high mortality or low ART coverage.

Why it matters

- Improves visibility of global and regional HIV burden.
- Highlights countries where low ART coverage may be contributing to higher mortality.
- Supports targeted resource allocation rather than equal but inefficient distribution.
- Provides executive KPIs for monitoring treatment effectiveness year by year.

Recommendations

- Prioritise high-burden countries with low ART coverage, especially Nigeria and other countries showing both high infection volume and treatment gaps.
- Use ART coverage as a core public health performance indicator, with an 80% minimum coverage target for sustained mortality reduction.
- Increase prevention investment in countries where new infections remain high despite treatment progress.
- Build routine risk-monitoring dashboards that combine infection volume, deaths, ART coverage and mortality rate into a single executive view.
- Use country-level dashboards to support policy communication, donor prioritisation and programme evaluation.

Author

Olufunmilola Lanade

Healthcare Data Analyst Enthusiast

Power BI | DAX | Power Query | Healthcare Analytics | Data Storytelling